Simulation of a QPSK-Based Satellite Downlink with Link Budget Analysis and BER Performance Evaluation

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***Abstract*—Satellite downlink communication faces major challenges, including severe path loss over long distances, signal disruption from rain and atmospheric gases, and spectral leakage caused by sharp signal pulses. This paper presents a complete physical layer simulation of a satellite communication system built using GNU Octave. The simulation begins by generating digital bits and modulating them using the QPSK modulation scheme. To prevent spectral leakage, an RRC filter is applied. The signal is then passed through a simulated channel that accounts for a 36,000 km geostationary satellite distance, along with fixed rain fade and antenna losses. At the ground station, thermal noise generated within the receiver's own circuitry is added, modeled as Additive White Gaussian Noise(AWGN) and a matched RRC filter is used to recover the signal before decoding. Path loss is addressed through received antenna gain rather than active amplification. System performance is validated both at a single operating point of 5.42 dB Eb/N0 and across a systematic sweep from 0 to 14 dB, comparing simulated Bit Error Rate(BER) against theoretical benchmarks at each point. The results confirm that the receiver can accurately recover the transmitted data, with the simulated BER closely following the theoretical expectations throughout the tested range.**

***Index Terms*—Satellite communication, QPSK modulation, Root Raised Cosine(RRC) filter, Bit Error Rate(BER), path loss, Additive White Gaussian Noise(AWGN)**

I. Introduction

This paper presents a complete satellite communication simulation built in GNU Octave(MATLAB-like syntax). The simulation begins by generating digital bits and modulating them using QPSK modulation, where we get two components, the I and Q components, which are then combined together to form a physical wave. This physical wave is a square-shaped wave and since this wave exists across all frequencies, it may disrupt other frequency channels if left unfiltered. To prevent this, the wave is smoothened using an RRC filter at the transmitter before it is sent out. The wave is then sent out from the satellite towards the receiver station on Earth and in between lies the channel, which includes both the vacuum of space and Earth's atmosphere. The large distance from satellite to earth is responsible for the path loss of the signal, since the signal tends to spread out as it leaves the satellite. After entering Earth's atmosphere, the signal is further affected by rain, which causes attenuation, by gases that absorb part of the signal and by particles present in the atmosphere that deflect the signal's path, reducing how much of it actually reaches the antenna. When the signal finally reaches the receiver station, the random thermal agitation of electrons within the receiver's own circuitry generates thermal noise, which adds to the noise the signal already experienced. This thermal noise is modeled in the simulation as Additive White Gaussian Noise(AWGN). It is considered only at the receiver and not the transmitter, since the transmitted signal is strong enough that the transmitter's own thermal noise is negligible in comparison; by the time the signal reaches the receiver, however, it has weakened considerably and the receiver's thermal noise becomes significant relative to it. There is a matched RRC filter at the receiver that filters out this added noise and converts the physical waveform back to digital bits. For the problem of path loss, a large receiving antenna is used, since a larger antenna captures more of the incoming signal through its gain, rather than through active amplification. The system's performance is validated both at a single operating point of 5.42 dB Eb/N0 and across a systematic sweep from 0 to 14 dB, comparing simulated Bit Error Rate against theoretical benchmarks at each point, confirming that the receiver correctly recovers the original transmitted data.

II. The Transmitter

A geostationary satellite is considered, positioned 36,000 km above the Earth's equator. The data to be transmitted is generated using the randi() function in Octave/MATLAB, producing a random stream of 1000 digital bits. These bits are paired together and mapped to QPSK symbols, with each pair of bits producing one symbol. The bit pairs are directly mapped to In-phase(I) and Quadrature(Q) values: a bit value of 0 maps to +1 and a bit value of 1 maps to -1, giving four possible symbol combinations - (1,1), (-1,1), (1,-1), (-1,-1) - corresponding to the four phase states of QPSK. Each symbol is then upsampled by a factor equal to the number of samples per symbol, preparing the I and Q streams for pulse shaping. A Root Raised Cosine (RRC) filter, designed with a roll-off factor(beta) of 0.35 and a span of 6 symbols, is

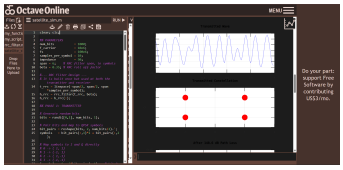


Fig. 1. Transmitted waveform and constellation diagram

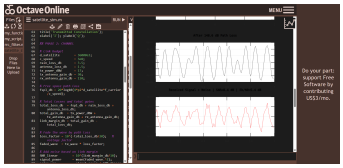


Fig. 2. Path loss and received signal

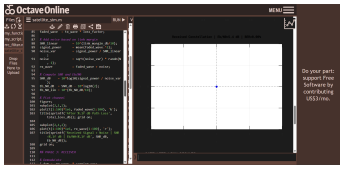


Fig. 3. Received constellation

applied to both the I and Q streams through convolution. This is the pulse-shaping step that smooths the sharp edges of the signal before transmission and the same filter is later reused at the receiver as a matched filter. The shaped I and Q signals are then modulated onto a 10 MHz carrier: the I component modulates a cosine carrier and the Q component modulates a sine carrier, and the two are combined to produce the final transmitted waveform. The resulting signal, along with the original transmitted constellation(the I and Q symbol values before pulse shaping and modulation), is plotted to visually confirm the correct signal generation before the waveform is passed into the channel.

III. The Channel

Once transmitted, the waveform must traverse the physical channel separating the satellite from its ground station - a channel that introduces several distinct sources of signal degradation, each of which is modeled explicitly in this simulation. The dominant loss mechanism is free-space path loss(FSPL), calculated using the standard formula relating loss to the transmission distance, carrier frequency and the speed of light. Given the 36,000 km link distance and a 10 Mhz carrier, this loss is substantial, reflecting how significantly a signal's power density diminishes as it spreads outward over such a distance. To this, two additional fixed losses are added: a 3.5 dB allowance for rain fade and a 1.5 dB allowance for antenna inefficiency - both treated as constant values representative of typical conditions, rather than as probabilistic or time-varying effects. The link margin then determines how the channel is simulated numerically. On the gain side of the link budget, three contributions are summed: the transmit power of the satellite(17 dBW), the transmit antenna gain(30 dB) and the receive antenna gain(110 dB). The difference between this total gain and the total loss yields the link margin - a single figure representing how much surplus signal strength remains once every loss has been accounted for. The transmitted waveform is first scaled down according to the total loss, producing the faded signal. Gaussian noise is then generated with a variance derived directly from the link margin(via the implied SNR) and added to this faded signal, producing the final received waveform. Because noise power is computed from the link margin rather than fixed independently, the resulting SNR and Eb/N0 values are an emergent property of the link budget itself, rather than parameters chosen in advance - allowing the system's operating point to reflect the physical assumptions built into the channel model.

IV. The Receiver

At the ground station, the receiver's task is to reverse the effects introduced during transmission and recover the original bit stream from a signal that has been attenuated, filtered and corrupted by noise. The first step is demodulation: the received waveform is multiplied separately by the same cosine and sine carrier references used at the transmitter, isolating the base band I and Q components once again. These demodulated signals still contain residual high-frequency content and accumulated noise, so they are passed through the RRC filter which now operates as a matched filter. Applying an identical filter response at both ends of the link is a deliberate design choice: it maximizes the signal-to-noise ratio at the sampling instant and completes the pulse-shaping without further widening the signal's bandwidth. Following the matched filtering, the continuous I and Q waveforms must be reduced back to discrete symbol values. This is done by sampling at a fixed offset - the midpoint of each symbol period - selecting one representative I and Q value per transmitted symbol. A straightforward sign-based decision rule is then applied to recover bits: a positive sampled value is interpreted as a transmitted bit of 0 and a negative value as a transmitted bit of 1, mirroring exactly the mapping convention used at the transmitter. The recovered bit sequence is compared directly against the original transmitted bits to count errors and compute the simulated Bit Error Rate for that run. To visualize the cumulative effect of channel loss and noise on the signal, the sampled I-Q values are plotted as a received constellation diagram, annotated with the corresponding Eb/N0 and BER for that operating point - allowing the spread and clustering of the recovered symbols to be inspected directly rather than inferred only from the error count.

V. Performance Analysis

The performance of the simulated satellite link was evaluated in two complementary ways: a single end-to-end validation run and a systematic sweep across a range of Eb/N0 values. In the single-run validation, the system was simulated at its naturally occurring operating point of 5.42 dB Eb/N0,

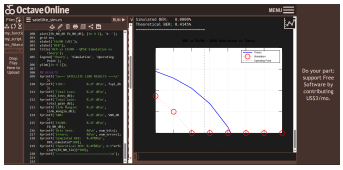


Fig. 4. Performance Analysis

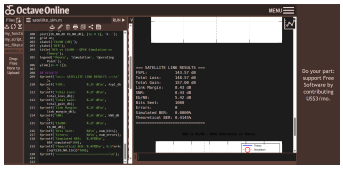


Fig. 5. Satellite Link Result

determined by the link budget described in the Channel section(a free-space path loss of 143.57 dB, combined with the fixed rain and antenna losses, against a total gain of 157.00 dB, yielding a link margin of 8.43 dB). At this operating point, 1000 bits were transmitted and 0 bit errors were observed, giving a simulated BER of 0.0000 percent compared to a theoretical BER benchmark of 0.4145 percent(evaluated per independent quadrature channel) at the same Eb/N0. While this result is consistent with the theoretical prediction, it should be interpreted cautiously: at this Eb/N0, a bit error probability of roughly 0.4 percent implies an expected 4 errors per 1000 bits, so observing 0 errors is a plausible outcome of natural statistical variation rather than strong confirmation that the simulated and theoretical BER match exactly. To evaluate performance more rigorously, a second test swept the Eb/N0 value from 0 to 14 dB in 2 dB increments, independently simulating 2000 bits at each point and comparing the resulting simulated BER against the theoretical QPSK BER curve. As expected, BER decreased sharply as Eb/N0 increased in both the theoretical and simulated results, with the simulated points closely tracking the theoretical curve, particularly at lower Eb/N0 values where bit errors occur frequently enough to be statistically meaningful within a 2000-bit sample. At higher Eb/N0 values, where the true BER falls below what 2000 bits can reliably resolve, simulated points naturally reached zero errors; these are plotted at a floor value of 1e-4 purely to remain visible on logarithmic scale, and should be read as ”no errors observed in this sample” rather than a precise BER measurement. Taken together, these results confirm that the transmitter, channel and receiver stages of the simulation are correctly implemented: the recovered bit stream matches the theoretical performance expected of a QPSK system operating under the modeled path loss, rain fade and thermal noise conditions.

VI. Conclusion

This paper presented a complete physical-layer simulation of a satellite downlink communication system, built in GNU Octave, covering the full signal chain from bit generation to signal recovery. QPSK modulation was used to convert digital bits into a physical waveform, with a Root Raised Cosine (RRC) filter applied at the transmitter to prevent spectral leakage caused by the signal’s naturally square-shaped edges. The simulated channel modeled a 36,000 km geostationary satellite link, incorporating free-space path loss, fixed rain fade and antenna losses, and additive thermal noise at the receiver, modeled as Additive White Gaussian Noise (AWGN). A matched RRC filter and sign based decision logic at the receiver were used to recover the original transmitted bits. System performance was validated both at a single operating point of 5.42 dB Eb/N0 and throughout a systematic sweep from 0 to 14 dB, comparing the simulated BER against theoretical benchmarks at each point. The results confirm that the receiver is able to recover the transmitted data, with simulated BER closely tracking theoretical expectations across the tested range. Future work includes extending this simulation to incorporate a jamming signal and an adaptive LMS notch filter to study recovery under active interference.